Roll No. ....

4th Sem / Civil

Subject:- Soil Mechanics and Foundation Engineering

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 The type of soil formed by the deposition of sediments by rivers is (CO1)

a) Black soil b) Alluvial soil
c) Red soil d) Laterite soil

Q.2 The unit of permeability is (CO4)

a) m/s b) m/s^2
c) N/m^2 d) N-m

Q.3 Specific gravity of soil solids is measured using (CO2)

a) Pycnometer b) Hydrometer
c) Core cutter d) Thermometer

Q.4 Darcy's Law is related to (CO4)

a) Permeability b) Compaction
c) Consolidation d) Shear strength

Q.5 Which one is a fine-grained soil? (CO3)

a) Sand b) Clay
c) Gravel d) Boulder

Q.6 The ratio of volume of voids total volume is (CO2)

a) Porosity b) Saturation

c) Void ratio d) Water content

Q.7 The instrument used to determine liquid limit of soil is (CO3)

a) Permeameter b) Casagrande apparatus

c) Core cutter d) Hydrometer

Q.8 The term cohesion is associated with (CO7)

a) Gravel b) Sand

c) Silt d) Clay

Q.9 Which method is used to determine in-situ density of soil? (CO2)

a) Sand replacement b) Liquid limit

c) Oven drying d) Hydrometer

Q.10 A well-graded soil contains (CO3)

a) Particles of one size

b) Mostly clay

c) a good range of all sizes

d) Organic content

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Alluvial soil is found mostly in________ areas. (CO1)

Q.12 ______ test is used to determine permeability.(CO4)

Q.13 The ratio of weight of water to weight of solids in soil is______. (CO2)

Q.14 The apparatus used for determining dry density is______. (CO2)

Q.15 Clayey soil has high permeability.(True/False) (CO4)

Q.16 Direct shear test is used for cohesion-less soils. (True/False) (CO7)

Q.17 _____ is the moisture content at which soil changes from plastic to liquid state. (CO3)

Q.18 Water table affects the effective stress of soil. (True/False) (CO5)

Q.19 ______ foundation is used when hard strata is deep below. (CO10)

Q.20 Name any one method of soil stabilization. (CO10)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Define alluvial soil. State two engineering properties. (CO1)

Q.22 Define total stress, effective stress and neutral stress acting on soil mass. (CO5)

Q.23 Explain the different methods of field compaction. (CO8)

Q.24 Define pile. Give the necessity of pile. (CO11)

Q.25 State differences between compaction and consolidation. (Any five) (CO6)

Q.26 What is liquid limit? How is it determined? (CO3)

Q.27 A soil sample has bulk density = 2.65 gm /cc and water content = 15%. Find dry density. (CO2)

Q.28 Define Darcy's Law. What is coefficient of permeability? (CO4)

Q.29 Define isobar and pressure bulb. (CO10)

Q.30 What is cohesion? Explain with an example. (CO7)

Q.31 Define ultimate bearing capacity, Net ultimate bearing capacity. (CO10)

Q.32 Define area ratio and recovery ratio. State its significance. (CO9)

Q.33 Differentiate between isolated footing and combined footing. (Any five) (CO11)

Q.34 Explain methods of soil exploration. (CO9)

Q.35 Define soil stabilization. Name any two commonly used techniques. (CO10)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain different types of shallow and deep foundations. (CO11)

Q.37 Find the shear strength of the soil if the cohesion (C) = 53 kpa, then angle of internal friction (ϕ) =24^o. The normal stress on the soil is 125 kpa. (CO7)

Q.38 Explain Wash and Precussion boring. State their advantages and disadvantages. (CO9)